

Vesicular–arbuscular mycorrhizal infections in root organ cultures

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Vesicular–arbuscular mycorrhizal infections, similar to those formed in plants with leaves, have been established for the first time in clover root organ cultures. These infections are most easily established in roots grown on a modified White's agar medium but they can also form in liquid culture. The pH of the medium is a critical factor determining the initiation and spread of infection. White's medium inhibited spore germination and growing roots did not relieve this inhibition. Pre-germinated spores were therefore used as inoculum. The possible value of root organ cultures for the study of root : endophyte interactions is discussed in relation to studies of some obligate fungi.

INTRODUCTION

The *Endogone* species that form vesicular–arbuscular (VA) mycorrhiza in many plant roots are obligate symbionts. Typical infections can be produced in clover seedlings grown in agar culture [8]. In later experiments effects of calcium phytate and of monocalcium phosphate were studied [10]. Infection was reduced by increasing amounts of monocalcium phosphate in the medium; phytate, inositol and glycerol tended to increase the development of the external mycelium. These results are, however, difficult to interpret in terms of fungal physiology, because one cannot separate effects of the treatment on host metabolism from direct effects on the fungus. With plants grown in test-tubes manipulation and observation are difficult and roots are exposed to the light. Also, one cannot harvest the external mycelium without melting the agar and thus killing the mycelium. Root organ cultures therefore seemed to offer marked advantages and we investigated the possibility of obtaining VA mycorrhiza in such cultures, hoping that this might lead to dual cultures in liquid medium.

Typical VA infections produced in root organ cultures are described in this paper. It was at first difficult to obtain consistent results and we tried several approaches including divided plates, liquid culture, different methods of sterilizing the medium, different forms of phosphate, different pH of the medium, residual effects of roots on germ tube growth and effects of the medium on spore germination. Results of some of these tests are also described here.

MATERIALS AND METHODS

General

Glassware was soaked in chromic acid and thoroughly rinsed before use. AR grade chemicals were used where available and all media were prepared using double distilled water.

Media

For establishment and routine maintenance of root cultures a modified White's medium [17] of the following composition was used: in 1 l of water—KCl, 65 mg; KNO_3 , 80 mg; $\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$, 300 mg; $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, 720 mg; $\text{NaH}_2\text{PO}_4 \cdot 2\text{H}_2\text{O}$, 10.7 mg; FeNaEDTA, 4.6 mg; $\text{MnCl}_2 \cdot 4\text{H}_2\text{O}$, 4.9 mg; KI, 0.75 mg; H_3BO_3 , 1.5 mg; $\text{ZnSO}_4 \cdot \text{H}_2\text{O}$, 1.92 mg; $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, 1 μg ; $\text{Na}_2\text{MoO}_4 \cdot 2\text{H}_2\text{O}$, 0.168 μg ; glycine, 3 mg; thiamine HCl, 0.1 mg; nicotinic acid, 0.5 mg; pyridoxine, 0.1 mg; sucrose, 20 g. The pH of this medium was adjusted to 4.9 before sterilization at 121 °C for 15 min.

This medium was modified for establishment of mycorrhizal infections by omitting the sodium dihydrogen phosphate, replacing it by either 9.3 mg potassium phosphate (KH_2PO_4) or 10.7 mg calcium phosphate ($\text{CaHPO}_4 \cdot 2\text{H}_2\text{O}$) and adjusting the initial pH to 7.0. This medium was sterilized in a pressure cooker at 121 °C for 10 min.

Unexpectedly no infections developed in media which had been sterilized in a large, steam injection autoclave, though root growth was normal. Subsequently media used for the establishment of mycorrhiza were always sterilized in a domestic pressure cooker which cools down within 10 to 15 min compared to 30 to 45 min in the large autoclave. This observation seems worth recording in case others may encounter similar difficulties.

Divided plates were prepared using modified White's agar medium containing calcium phosphate (pH 4.9) on one side and a medium which was found to be suitable for the infection of whole plants [10] on the other side. This medium has the following composition: in 1 l of water— KNO_3 , 500 mg; $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, 200 mg; CaCl_2 , 200 mg; FeNaEDTA, 46 mg; calcium phytate, 630 mg; Difco agar, 10 g, pH 6.8.

Establishment and maintenance of root cultures

Seeds of red clover, *Trifolium pratense* (cultivar S123) and *T. parviflorum*, were sterilized in concentrated sulphuric acid for 20 and 15 min respectively, washed eight times with sterile distilled water and germinated in the dark on inverted agar plates. Seeds of tomato, *Lycopersicon esculentum* (Moneymaker) were sterilized by shaking in undiluted "Chlorox" (I.C.I. Ltd.) for 20 min and after washing were germinated on damp filter paper. When the radicals were approximately 10 mm long, 8 mm root tips were excised and transferred to 150 ml Erlenmeyer flasks containing 50 ml of modified White's medium. In the case of *T. pratense* and *T. parviflorum* 10 root tips were grown together in the same flask [13]. Continuous root cultures were established from those root tips showing good growth, by subculturing 30 mm of the apical portion into fresh medium every 10 days. Stock root cultures were maintained at 27°C in the dark.

Establishment of mycorrhizas in excised roots

A 20-mm length of the main root axis bearing several laterals was placed in a Petri dish containing either 10 ml of White's modified medium or 30 ml of medium solidified with 8 g/l of Difco agar. Yellow vacuolate spores excised from sporocarps of *E. mosseae* [11], stored at 6 °C for several months, were used as inoculum. The spores were sterilized with streptomycin and chloramine T [8] and germinated in

groups of five or six on 1% agar plates. After 5 to 7 days, groups of uncontaminated, germinated spores were removed on small pieces of agar and placed close to the roots. The inoculated roots which were maintained at 18 to 20 °C in the dark, were examined periodically and stained with lactophenol aniline blue [10] after 21 to 28 days' growth.

RESULTS

Root growth

Single root tips taken from seedlings of *T. pratense* and *T. parviflorum* usually failed to grow when transferred to liquid medium, but when cultures were initiated by placing 10 root tips in each flask, several started to grow and could be sub-cultured. Roots of *T. parviflorum* in liquid culture grew more slowly than those of *T. pratense*, showed an initial growth lag and became rather brown, before producing many laterals. Most of the experiments described were done with one clone of *T. pratense* which, during the period of about one year, was sub-cultured 32 times. A few cultures were also set up using *T. parviflorum* roots on their third passage.

Description of infections

Of 163 cultures 53 became mycorrhizal. As they were experimental, not all the cultures were expected to become infected. There was some variability between groups of germinated spores placed near different parts of the same root system in the same Petri dish. Successful infection seemed to depend more on condition and proximity of nearby roots than on number or size of spores in the inoculum. Freshly germinated spores were preferable to older ones with germ tubes in which septa had already formed. Four different kinds of development could follow inoculation: (1) Germ tubes became empty and septate and spores moribund. (2) Germ tubes made quite extensive new growth and formed new vegetative spores, but roots and inoculum remained quite separate. (3) After vigorous growth of germ tubes, hyphae became closely adpressed to the root surface, often forming a pad of mycelium almost resembling a localized mantle. Some hyphal tips became firmly attached to the root surface, usually resulting in a diffuse blue staining reaction where penetration may have been attempted, but no appressorium developed and there was no penetration of the root. (4) Appressoria formed, followed by successful penetration of the root and a marked growth stimulation of the external mycelium. Most extensive infection occurred when germ tubes extended rapidly following inoculation and encountered young lateral roots, but there was little or no apparent attraction of germ tubes towards the roots until they approached them closely. Many infections started 0.5 to 1 cm behind the root tip. In the most successful cultures up to half the root system became mycorrhizal. Mycorrhizal infections rarely developed if root growth was poor or roots had turned brown.

There were no observable changes in root morphology following infection and one could not identify infected roots until they were stained. As well as in *T. pratense*, normal VA infections developed in cultures of *T. parviflorum*.

Successful infections were in every way typical and similar to those of plants with leaves. Appressoria were formed (Plate 1) and at entry points, penetrating hyphae

characteristically stood out from the root surface at right angles (Plates 2 and 4). This could be the result of continued sub-apical growth after the hyphal tip has become firmly attached to the root surface. When infections occurred early, well-developed arbuscules usually formed (Plates 2 and 3), but when they developed rather late in root cultures already quite old, the arbuscules often remained vestigial and looked stumpy with few fine branches. In such infections the main feature was massive, much swollen hyphae growing longitudinally within the intercellular spaces of the cortex towards the root apex. Vesicle development was less affected by the age of the root culture. As on whole plants, vesicles developed later than the arbuscules. Plate

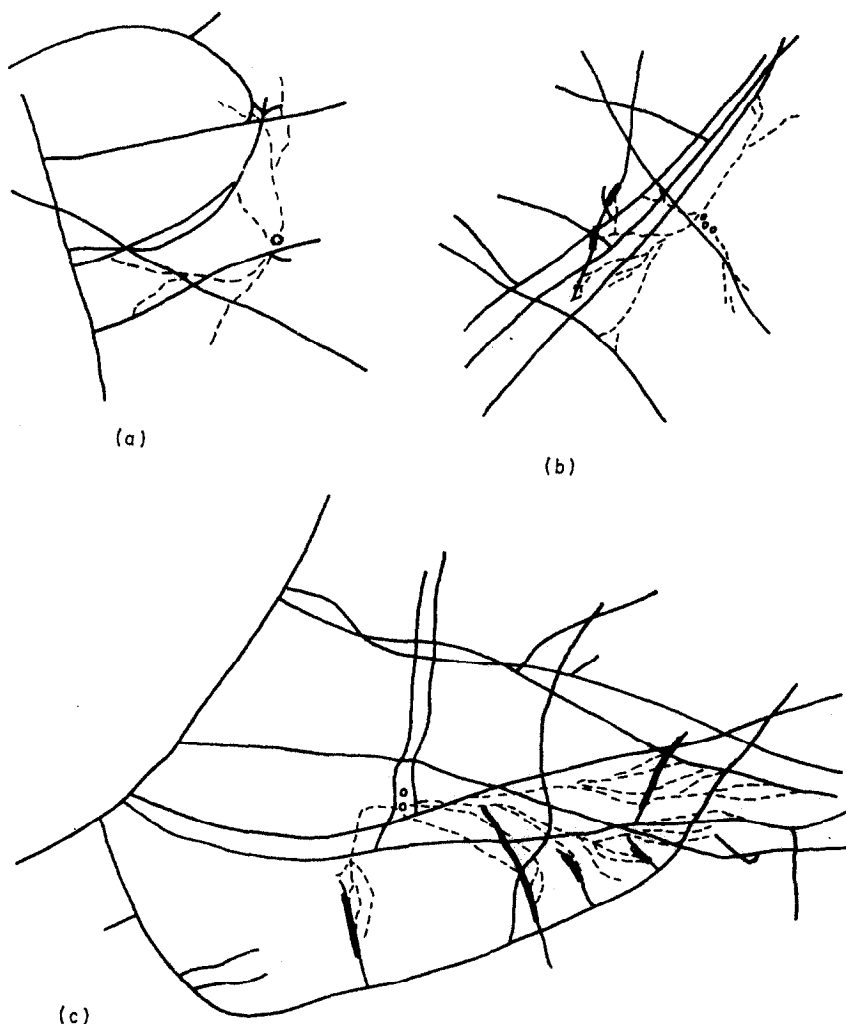


FIG. 1. Drawings of stages in the establishment of mycorrhizal infection in root organ cultures of red clover. Magn. approx. $\times 2\frac{1}{2}$. Note the extent of germ tube growth without infection in (a). (b) Shows the beginning and (c) the spread of an infection. Note that infections occur at some distance from the point of inoculation. (—) Roots; (—) infected roots; (---) hyphae; (O) spore inoculum.

PLATES 1-10. Stages in the development of vesicular-arbuscular infection and external mycelium in root organ cultures of red clover infected with *E. mosseae*. Scale represents 100 μ .

PLATE 1. Section through two appressoria.

PLATE 2. Young infection with entry point and arbuscules.

PLATE 3. Section showing details of arbuscules.

PLATE 4. Well-infected older root with prominent entry points and many vesicles.

PLATE 5. Root with attached external mycelium bearing newly formed vegetative spores. Two yellow vacuolate spores of the inoculum are marked by arrows.

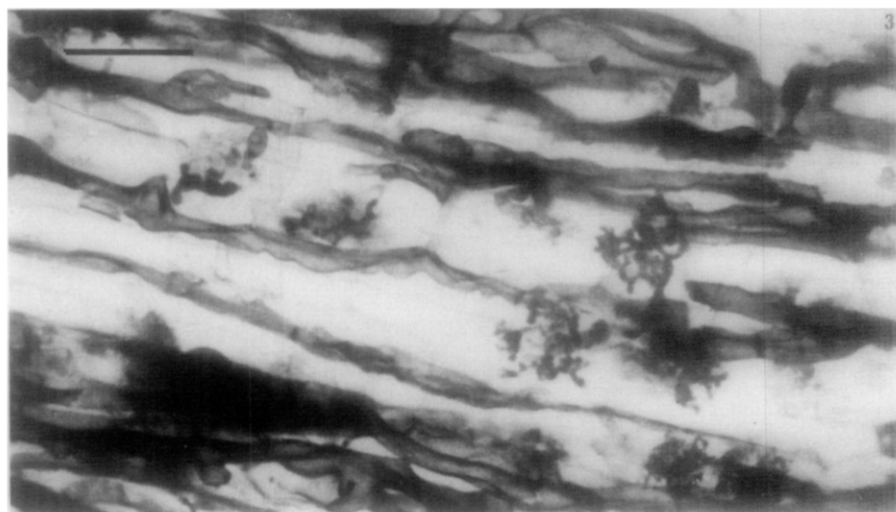
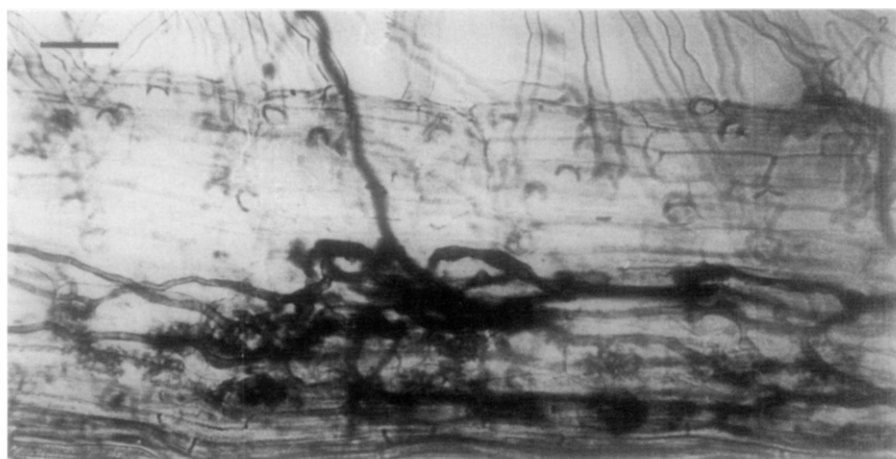
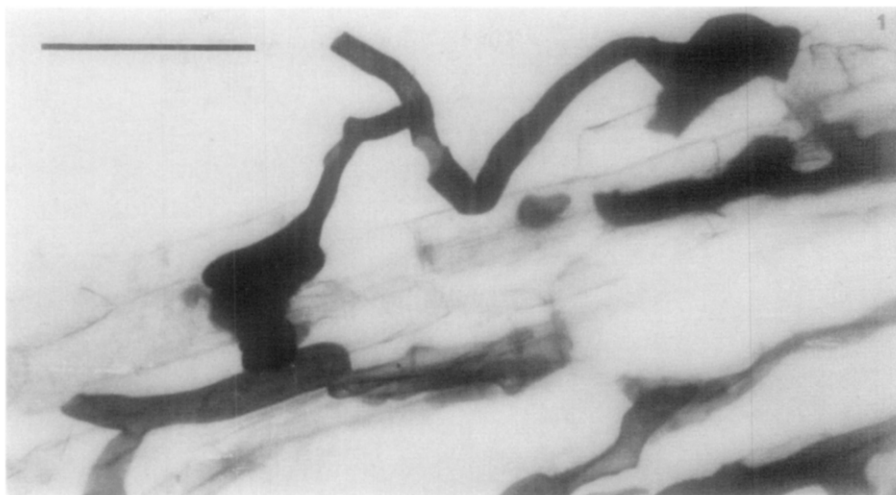
PLATE 6. External mycelium growing across root bearing root hairs. Arrows mark calcium phytate particles.

PLATE 7. External mycelium with "arbuscule-like" branches.

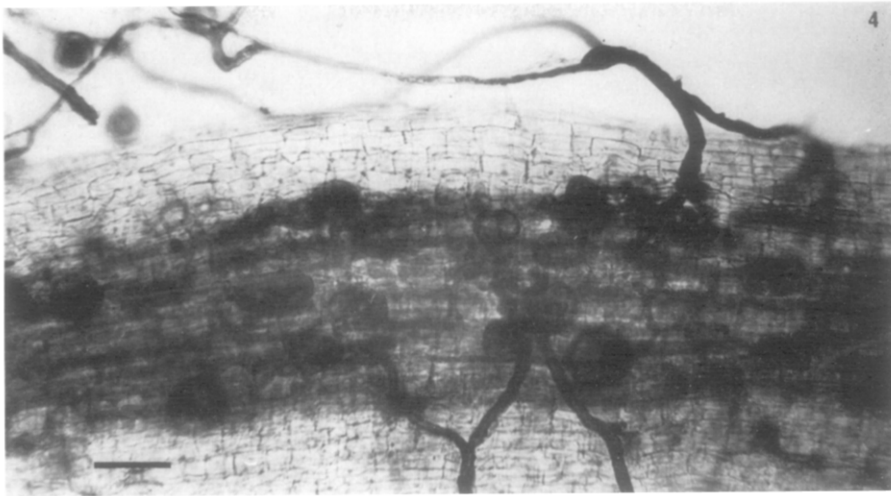
PLATE 8. External mycelium with branches and a vegetative spore.

PLATE 9. "Sympodial" growth of hypha bearing a vegetative spore.

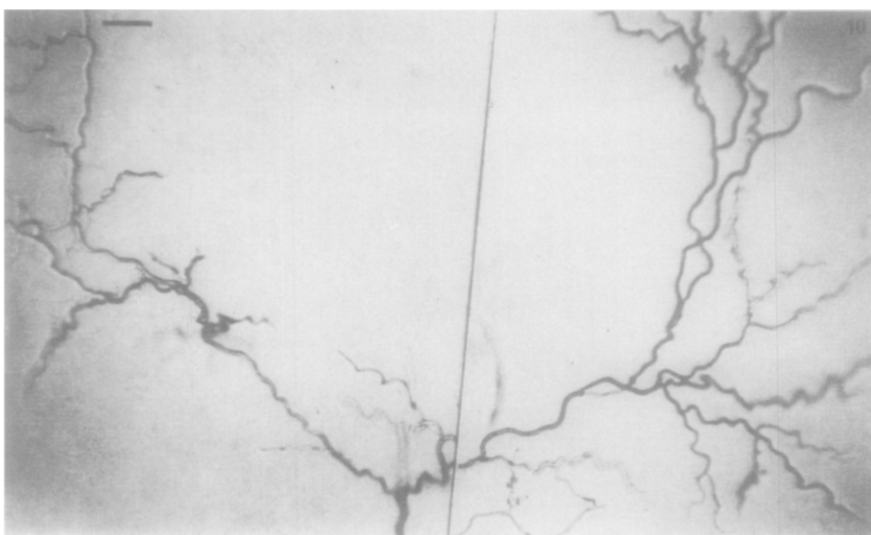
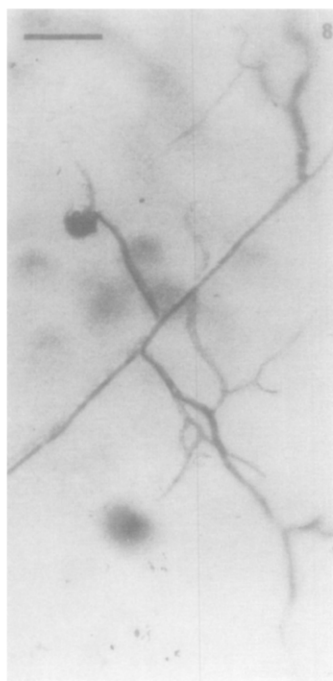
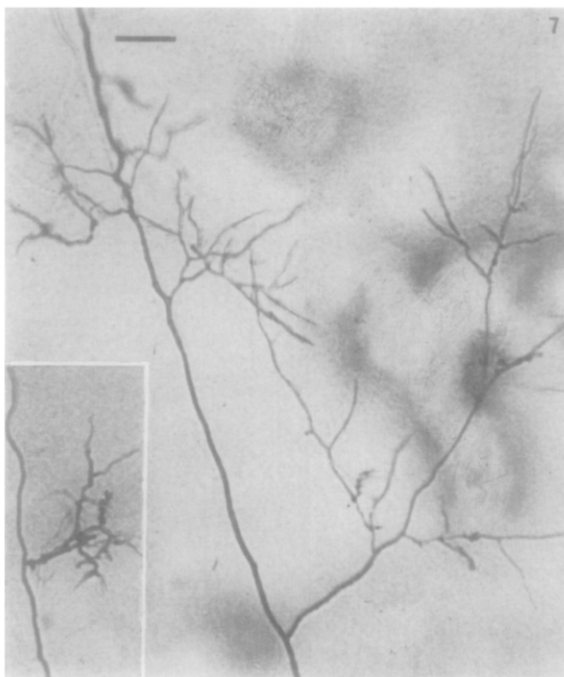
PLATE 10. "Spiral" growth of external mycelium.



PLATES 1-3



PLATES 4-6



PLATES 7-10

4 shows a well-developed infection with vesicles. Most of the infection is in the inner cortical layers.

After infection the external mycelium always grew vigorously, branching quite profusely and producing many small, round vegetative spores (Plates 4 and 5). These arose terminally on short lateral branches, but quite often the hyphal tip resumed growth after spore formation suggesting a sympodial form of growth (Plate 9). Such spores, when detached, could give the mistaken impression that they had arisen from two separate subtending hyphae. Hyphae extended up to 30 to 40 mm from the root surface. Plate 6 gives some impression of this and also shows that the hyphae were neither attached to nor deflected towards particles of calcium phytate in the medium, although this often increased infection. Most interesting was a form of branching strongly reminiscent of arbuscules (Plates 7 and 8) which occurred extensively in some successful cultures. It appeared to be a localized response to an unknown stimulus but generally occurred throughout some plates. In soil, remnants of similar fine, much-branched, rhizoid-like structures are also found, often attached to particles of organic matter. In agar cultures such branching appeared to arise without attachment to a particulate surface. Another reaction to external stimuli was a growth form that resulted in very wavy hyphae (Plate 10) sometimes distinctly spiral in origin.

In spite of its extensive and vigorous growth it was not possible to sub-culture from the external mycelium and even if left *in situ* it grew little when separated from the infected root.

Location of infection

In several cultures we traced the location of the first infections in relation to the inoculum. Figure 1 shows scale drawings of root configurations and sites of infection in relation to hyphal growth from the inoculum. Figure 1 (a) illustrates the extensive germ tube growth from a single spore; no infection had yet occurred; (b) shows the position of the first infection in relation to an inoculum consisting of three spores; (c) shows a well-established infection with five separate primary infection sites. In none of 15 mapped samples did infection occur close to the site of inoculation or in the first roots encountered by the germ tubes. Generally first infections were 5 mm or more removed from the site of inoculation which was always close to a root. Recently developed lateral roots appeared to be preferred infection sites. Once an infection was established in a root, it seemed to become more prone to further infection from new entry points.

pH of the medium

Of all variables examined, pH of the medium appeared to be the most important factor determining success or failure. The pH of all media containing agar changed during sterilization; those adjusted to 3.4 and 4.9 rose to 4.5 and 5.2 respectively, while the one adjusted to pH 7.0 fell to 6.7. The pH of the agar was also changed by the roots during the experiments. Those cultures starting at pH 6.7 usually rose to about pH 8 and then dropped rapidly to between pH 4 and 5 but the actual rate at which this occurred varied between plates. Cultures starting at a lower pH changed only slightly during the experiment. When the initial pH of the agar was 4.5, the growth of the fungus was inhibited and although the roots grew well they did not

become infected at all (Table 1). Presumably if the fungus fails to get established and enter the root during the period when the pH is favourable, then inconsistent results are obtained.

TABLE 1
The effect of pH on the hyphal growth in the agar and on the number of root infections

Initial pH of medium	No. of replicates	Total no. of spores in growth categories				Total no. of root infections
		Extensive	Moderate	Slight	Negligible	
6.7	21	59	6	15	15	16
4.5	28	8	8	83	53	0

We tried to buffer the medium with a phosphate buffer or by adding solid calcium carbonate. Because high phosphate concentrations in the plant tissue apparently inhibit infection [9, 10] we used a low concentration, 100 μ M potassium phosphate buffer at pH 6.8. This was insufficient to prevent the drop in pH. With solid calcium carbonate in the medium the pH remained between 7 and 8. Mycelium in the agar grew vigorously with many finely divided branches, but only two good infections developed from 10 inoculations, compared with two good infections from three inoculations without calcium carbonate. In the presence of the calcium carbonate some roots looked abnormal; it seemed that the calcium carbonate might have caused some changes in root metabolism that made the roots more resistant to infection.

Different forms of phosphate

In test-tube cultures with entire clover seedlings the most extensive infection occurred when the phosphate source was calcium phytate [10]. In root organ cultures also calcium phytate seemed to have some beneficial effects on infection, but good infections also occurred with potassium dihydrogen phosphate and with calcium phosphate.

Divided plates

In divided plates the effect of placing the inoculum on a medium lacking sugar, vitamins and glycine was observed. Roots grew quite well with their proximal end in the complete medium and good infections developed on the distal side in a mineral salts medium containing calcium phytate. Infections were, however, no more extensive than in a 1 : 1 mixture of the two media.

TABLE 2
Effect of plant roots on the germination of E. mosseae spores

Medium	% Germination at 14 days
Water agar	95
White's	12
White's + red clover roots	10
White's + tomato roots	5

Liquid culture

Liquid culture of roots was quite successful, but it was difficult to establish infections and only a few root systems became strongly infected. Often germ tubes grew along and around the root but failed to penetrate it. Further work is needed to achieve the aim of producing a good supply of external mycelium.

Spore germination

In four tests we examined the possibility of using sterilized spores directly as inoculum without previously germinating them. Very few spores germinated and no roots became infected.

We then tested spore germination on modified White's medium with and without roots. Results shown in Table 2 suggested that some component in White's medium inhibited germination, irrespective of whether roots were present or not. Germ tube growth of the few spores that germinated in White's medium was minimal. Branching of germ tubes was markedly stimulated when calcium phytate (0.063%), inositol (0.02%) or glycerol (0.006%) were added to the distilled water agar; glycerol also induced the formation of many small vegetative spores.

DISCUSSION

Work with obligate fungi poses special problems and the kind of studies described here are well-recognized methods of by-passing some of these problems. Ultimately one hopes to elucidate the nature of the host dependence. In the meantime root organ cultures will be very useful for studying the movement of substances between host and fungus. The apparent inability of the fungus to infect at low pH of the medium agrees well with the behaviour of this particular endophyte in low pH soils [5]. This, as well as the entirely normal appearance of infections in root organ cultures, support the view that they closely resemble those in entire plants. As some of the root organ cultures were at their 32nd transfer there cannot be any carry over of substances from cotyledons or hypocotyl and one may conclude that, provided normal root growth is maintained, the endophyte does not require metabolites synthesized specifically in the leaves. The particular mechanism restricting the endophyte to the primary cortex and preventing its uncontrolled spread in the root also functions independently of leaves.

In general the fungus emerges as the weaker partner of the association and root penetration appears critical. Our observations suggest two separate phases, an attachment to the root surface which may or may not be followed by root penetration. Attachment, even without penetration, provides some stimulus for germ tubes, and produces some response in the underlying host cells which stain more heavily. Similar reactions occur in the infection processes of many fungal pathogens. Sometimes an adhesive substance is secreted by the host [12] or by the fungus [18] and there is frequently a digestion of outer wall layers. At the same time cytoplasm aggregates in the underlying host cell, a papilla is formed on the inner side of the epidermal wall and this is penetrated by an infection peg [2, 3, 14, 15]. No EM pictures of the primary entry point of a VA endophyte are yet available but root organ cultures are well suited to provide material for such a study. The heavy staining of cells below

germ tube attachment points probably indicates a stimulation of host cytoplasm similar to that preceding intracellular penetration.

Not all roots are equally suitable for attachment and often germ tubes grew straight across roots without any apparent reaction to their presence. Young lateral roots seemed to exert most morphogenetic effects on the fungus, and such effects may be a pre-condition for infection by germ tubes. Morphogenetic effects of host diffusates also precede penetration by *Thanatephorus cucumeris* [4]. Already mycorrhizal roots are more prone to further penetration, either because the fungus is invigorated, because root metabolism is changed, or both. Changes in hormone pattern [6] and permeability of host cells [7] are known to result from some fungal infections.

Although VA endophytes are classified as symbionts they have many similarities to obligate haustorial parasites of the *Erysiphales*, *Peronosporales* and mucoraceous mycoparasites, the haustoria of which resemble mycorrhizal arbuscules. The suggestion has been made [1] that such structures may be necessary for uptake of nutrients and transfer of those plant metabolites upon which the obligate relationship rests. It is therefore interesting and promising for their eventual culture that the finely branched, thin-walled structures very similar to arbuscules can develop outside the root in the agar medium.

The synergistic effect on germination of several, as opposed to single, spores has also been observed in some rust fungi [15, 16], where it has been attributed to leakiness of the spores. Although *Endogone* spores have thick walls they may suffer from similar loss of nutrients.

The production of mycorrhiza in root organ cultures offers many new and interesting possibilities for their future study.

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REFERENCES

1. BARNETT, H. L. & BINDER, F. L. (1973). The fungal host-parasite relationship. *Annual Review of Phytopathology* **11**, 273-292.
2. BUSHNELL, W. R. (1971). The haustorium of *Erysiphe graminis*: an experimental study by light microscopy. In *Plant-Parasite Interaction*, Ed. by S. Akai and S. Ouchi, pp. 229-254. The Phytopathological Society of Japan, Tokyo.
3. EDWARDS, H. H. & ALLEN, P. J. (1970). A fine-structure study of the primary infection process during infection of barley by *Erysiphe graminis* f. sp. *hordei*. *Phytopathology* **60**, 1504-1509.
4. FLENTJE, N. T., DODMAN, R. L. & KERR, A. (1963). Mechanism of host penetration by *Thanatephorus cucumeris*. *Australian Journal of Biological Sciences* **16**, 784-799.
5. HAYMAN, D. S. & MOSSE, B. (1971). Plant growth responses to vesicular-arbuscular mycorrhizas. I. Growth of *Endogone*-inoculated plants in phosphate-deficient soils. *New Phytologist* **70**, 19-27.
6. HEITEFUSS, R. (1966). Nucleic acid metabolism in obligate parasitism. *Annual Review of Phytopathology* **4**, 221-244.
7. KECK, R. W. & HODGES, T. K. (1973). Membrane permeability in plants: changes induced by host-specific pathotoxins. *Phytopathology* **63**, 226-230.
8. MOSSE, B. (1962). The establishment of vesicular-arbuscular mycorrhiza under aseptic conditions. *Journal of General Microbiology* **27**, 509-520.
9. MOSSE, B. (1972). Growth of *Endogone* mycorrhiza in agar medium. *Rothamsted Experimental Station Report for 1971*. Part I, p. 93.
10. MOSSE, B. & PHILLIPS, J. M. (1971). The influence of phosphate and other nutrients on the development of vesicular-arbuscular mycorrhiza in culture. *Journal of General Microbiology* **69**, 157-166.
11. NICOLSON, T. H. & GERDEMANN, J. W. (1968). Mycorrhizal *Endogone* species. *Mycologia* **60**, 313-325.

12. PAUS, F. & RAA, J. (1973). An electron microscope study of infection and disease development in cucumber hypocotyls inoculated with *Cladosporium cucumerinum*. *Physiological Plant Pathology* **3**, 461–464.
13. ROBBINS, W. J. & HERVEY, A. (1969). Auxin and growth of excised roots of *Bryophyllum calycinum*. *Proceedings of the National Academy of Sciences of the U.S.A.* **64**, 495–497.
14. SARGENT, J. A., TOMMERUP, I. C. & INGRAM, D. S. (1973). The penetration of a susceptible lettuce variety by the downy mildew fungus *Bremia lactucae* Regel. *Physiological Plant Pathology* **3**, 231–239.
15. SCOTT, K. J. (1972). Obligate parasitism by phytopathogenic fungi. *Biological Reviews* **47**, 537–572.
16. SCOTT, K. J. & MACLEAN, D. J. (1969). Culturing of rust fungi. *Annual Review of Phytopathology* **7**, 123–146.
17. WHITE, P. R. (1954). *Cultivation of Animal and Plant Cells*, 2nd ed. The Ronald Press Company, New York.
18. WILLIAMS, P. H., AIST, J. R. & BHATTACHARYA, P. K. (1973). Host parasite relations in cabbage clubroot. In *Fungal Pathogenicity and the Plant's Response*, Ed. by R. J. W. Byrde and C. V. Cutting, pp. 141–155. Academic Press, London and New York.